

Comparison analysis of the original and deconvolved data in the autologous culture

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```
library(stringr)
library(tidyverse)
library(DESeq2)
library(limma)
library(ggplot2)
library(ggrepel)
library(ggpubr)
```

*Load Data

```
# Original autologous data
#data_auto <- read.table("../Results/Normalized_data/Autologous/unnormalized_counts.txt",
#                          header = TRUE, row.names = 1)
data_auto <- read.table("../Results/Normalized_data/Autologous/normalized_counts.txt",
                        header = TRUE, row.names = 1)
names(data_auto) <- str_replace(names(data_auto), "_auto", "_autologous")

# Monoculture data
#data_B <- read.table("../Results/Normalized_data/B_cell/unnormalized_counts.txt",
#                     header = TRUE, row.names = 1)
data_B <- read.table("../Results/Normalized_data/B_cell/normalized_counts.txt",
                     header = TRUE, row.names = 1)
names(data_B) <- str_replace(names(data_B), "_B", "_monoculture")

# Deconvolved autologous data
data_auto_deconv <- read.table("../GRN_revision_deconvolution/output/Expression_profiles/Expression_
                              header = TRUE, row.names = 1, sep = ",")
names(data_auto_deconv) <- str_replace(names(data_auto_deconv), "_auto", "_deconv")
names(data_auto_deconv) <- str_replace(names(data_auto_deconv), "J", "D")
```

**Merge Datasets

```
# Common genes
common_genes <- intersect(intersect(rownames(data_auto), rownames(data_B)), rownames(data_auto_deconv))

data_auto_filter <- data_auto[common_genes, , drop = FALSE] %>% drop_na()
data_B_filter <- data_B[common_genes, , drop = FALSE] %>% drop_na()
data_auto_deconv_filter <- data_auto_deconv[common_genes, , drop = FALSE] %>% drop_na()

combined_data <- cbind(data_auto_filter, data_B_filter, data_auto_deconv_filter)
```

*PCA Analysis

```

sample_pca <- function(input_data, Condition){
  pca.res <- prcomp(t(input_data))
  percentVar <- round(100 * pca.res$sdev ^ 2 / sum(pca.res$sdev ^ 2), 1)

  dataGG <- data.frame(PC1 = pca.res$x[, 1], PC2 = pca.res$x[, 2]) %>%
    rownames_to_column("Patient") %>%
    mutate(
      Day = str_split_fixed(colnames(input_data), pattern = "_", n = 4)[, 1] %>% as.factor(),
      Patient = str_split_fixed(colnames(input_data), pattern = "_", n = 4)[, 2],
      Replicate = str_split_fixed(colnames(input_data), pattern = "_", n = 4)[, 3],
      Culture = str_split_fixed(colnames(input_data), pattern = "_", n = 4)[, 4]
    )

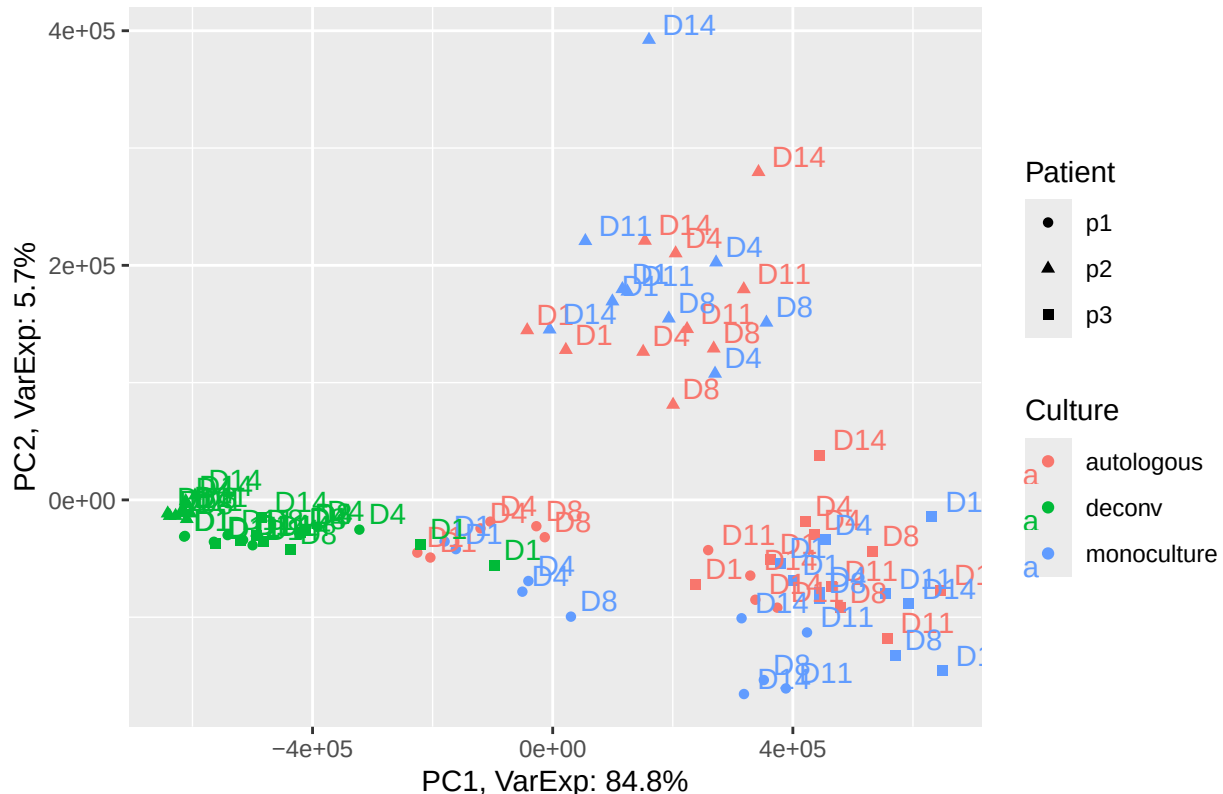
  dataGG$Day <- factor(dataGG$Day, levels = c("D1", "D4", "D8", "D11", "D14"))

  ggplot(dataGG, aes(PC1, PC2, col = Culture, shape = Patient)) +
    geom_point() +
    geom_text(aes(label = Day), hjust = -.25, vjust = -.25) +
    ggtitle(paste("PCA plot:", Condition, "culture")) +
    xlab(paste0("PC1, VarExp: ", percentVar[1], "%")) +
    ylab(paste0("PC2, VarExp: ", percentVar[2], "%")) +
    theme(plot.title = element_text(hjust = 0.5, size = 12, face = "bold"))
}

PCA_all_conditions <- sample_pca(combined_data, "auto-, mono- and deconvoluted auto-")
PCA_all_conditions

```

PCA plot: auto-, mono- and deconvoluted auto- culture



*Differential Gene Expression Analysis The analysis can be done if the un-normalised counts of the deconvoluted data are available, but it is not the case. We can perform the DGEA between the original autologous and monoculture, and in this case, we have to choose the un-normalised data in the previous chunk.

```
sample_info = as.data.frame(stringr::str_split_fixed(colnames(combined_data), pattern = "_", n = 4))
names(sample_info) <- c("Day", "Patient", "Replicate", "Condition")

ddsMat <- DESeq2::DESeqDataSetFromMatrix(countData = round(combined_data),
                                         colData = sample_info,
                                         design = ~ Condition)

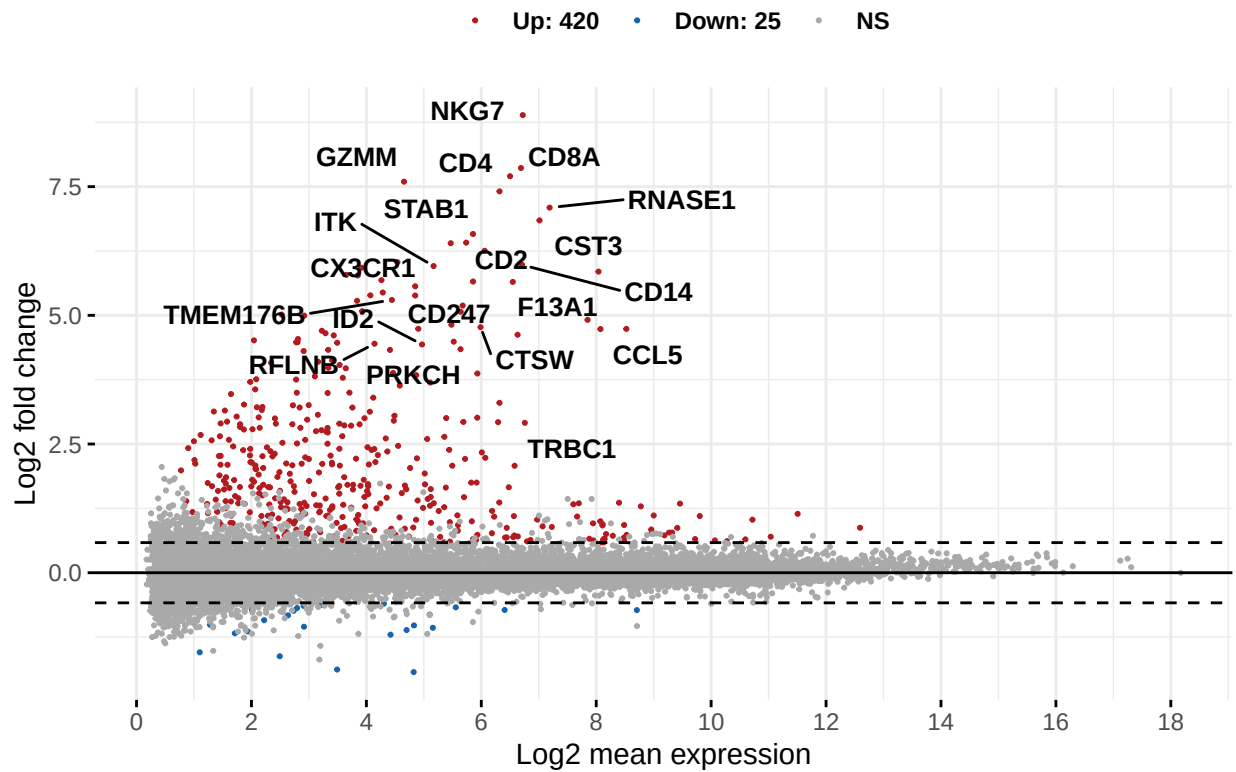
dds = DESeq(ddsMat)

contrast_condition <- c("Condition", "autologous", "monoculture")
results = results(dds, contrast = contrast_condition)
results_df <- as.data.frame(results)

ggma_plot <- ggmaplot(results_df,
                      main = "Autologous: original vs deconvoluted auto",
                      fdr = 0.05, fc = 1.5,
                      size = 0.4,
                      palette = c("#B31B21", "#1465AC", "darkgray"),
                      genenames = rownames(results_df),
                      legend = "top", top = 20,
                      font.label = c("bold", 11),
                      font.legend = "bold",
                      font.main = "bold",
                      ggtheme = ggplot2::theme_minimal())

ggma_plot
```

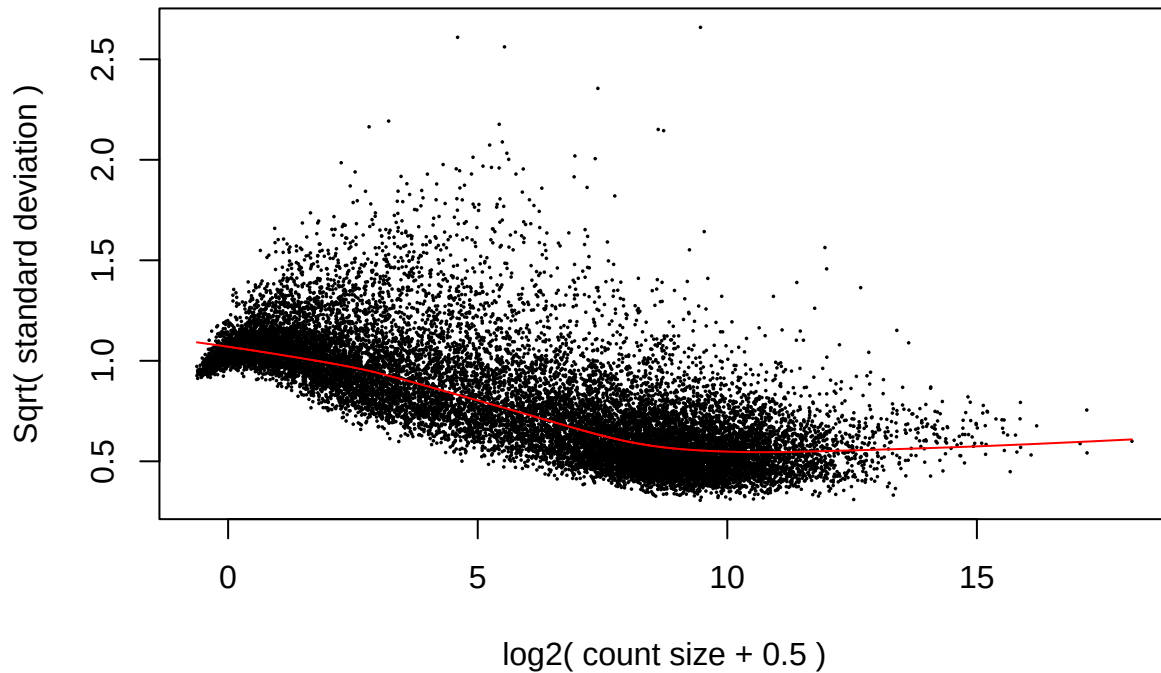
Autologous: original vs deconvoluted auto



```
sample_info$Condition <- factor(sample_info$Condition)
design <- model.matrix(~ 0+Condition, data = sample_info)
colnames(design) <- levels(sample_info$Condition)

v <- voom(combined_data, design, plot = TRUE)
```


voom: Mean–variance trend



```
fit <- lmFit(v, design)

contrast.matrix <- makeContrasts(autologous - deconv, levels = design)
fit2 <- contrasts.fit(fit, contrast.matrix)
fit2 <- eBayes(fit2)

results_limma <- topTable(fit2, sort.by = "p")

results_limma_df <- results_limma %>%
  dplyr::rename(
    baseMean = AveExpr,
    log2FoldChange = logFC,
    padj = adj.P.Val,
    pvalue = P.Value,
  )

# ggma_plot_limma <- ggmaplot(results_limma_df,
#                               main = "Original autologous vs deconvoluted",
#                               fdr = 0.05, fc = 1.5,
#                               size = 0.4,
#                               palette = c("#B31B21", "#1465AC", "darkgray"),
#                               genenames = rownames(results_limma_df),
#                               legend = "top", top = 20,
#                               font.label = c("bold", 11),
#                               font.legend = "bold",
```

```

# font.main = "bold",
# ggtheme = ggplot2::theme_minimal()
# ggma_plot_limma

*Correlation Analysis

gene_ids <- rep(common_genes, times = ncol(data_auto_filter))
vec1 <- as.vector(as.matrix(data_auto_filter))
vec2 <- as.vector(as.matrix(data_B_filter))
vec3 <- as.vector(as.matrix(data_auto_deconv_filter))

plot_df <- data.frame(
  Gene = gene_ids,
  Dataset1 = vec1,
  Dataset2 = vec2,
  Dataset3 = vec3
)

fit <- lm(Dataset1 ~ Dataset3, data = plot_df) #choose dataset
plot_df$residuals <- abs(resid(fit))

top_resid <- plot_df[order(-plot_df$residuals), ][1:10, ]
correlation <- cor(plot_df$Dataset1, plot_df$Dataset3, method = "spearman") #choose dataset

ggplot(plot_df, aes(x = Dataset1, y = Dataset3)) + #choose dataset
  geom_point(alpha = 0.4, color = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, color = "red", linetype = "dashed") +
  geom_point(data = top_resid, aes(x = Dataset1, y = Dataset3), #choose dataset
    color = "darkorange", size = 2) +
  geom_text_repel(data = top_resid, aes(label = Gene),
    color = "black", size = 3, max.overlaps = 20) +
  labs(
    title = paste("Gene expression correlation (r =", round(correlation, 3), ")"),
    x = "Original autologous",
    y = "Deconvoluted autologous"
  ) +
  theme_minimal()

```

Gene expression correlation (r = 0.955)

